



On Bayesian Analysis of Parsimonious Gaussian Mixture Models

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Abstract

Cluster analysis is the task of grouping a set of objects in such a way that objects in the same cluster are similar to each other. It is widely used in many fields including machine learning, bioinformatics, and computer graphics. In all of these applications, the partition is an inference goal, along with the number of clusters and their distinguishing characteristics. Mixtures of factor analyzers is a special case of model-based clustering which assumes the variance of each cluster comes from a factor analysis model. It simplifies the Gaussian mixture model through parameter dimension reduction and conceptually represents the variables as coming from a lower dimensional subspace where the clusters are separate. In this paper, we introduce a new RJMCMC (reversible-jump Markov chain Monte Carlo) inferential procedure for the family of constrained MFA models.

The three goals of inference here are the partition of the objects, estimation of the number of clusters, and identification and estimation of the covariance structure of the clusters; each therefore has posterior distributions. RJMCMC is the major sampling tool, which allows the dimension of the parameters to be estimated. We present simulations comparing the estimation of the clustering parameters and the partition between this inferential technique and previous methods. Finally, we illustrate these new methods with a dataset of DNA methylation measures for subjects with different brain tumor types. Our method uses four latent factors to correctly discover the five brain tumor types without assuming a constant variance structure and it classifies subjects with an excellent classification performance.

Keywords Mixture models · Factor analysis · Cluster analysis · Model-based clustering · RJMCMC · Bayesian clustering

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1 Introduction

Clustering models group a set of data objects into clusters where there are similarities within the same cluster and dissimilarities between different clusters. There are a number of clustering models and the Gaussian mixture model (GMM; McLachlan and Basford 1988) is one of the most popular one. It is a parametric probability density function represented as a weighted sum of Gaussian component densities:

$$p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{i=1}^m \omega_i f(\mathbf{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i),$$

where $\mathbf{X}$ is a continuous-valued data vector (i.e., measurement or features); m is the number of clusters; ω_i ($i = 1, \dots, m$) are the mixture weights with $\omega_i > 0$ and $\sum_{i=1}^m \omega_i = 1$; and $f(\mathbf{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ ($i = 1, \dots, m$) are the component Gaussian densities with mean vector $\boldsymbol{\mu}_i$ and covariance matrix $\boldsymbol{\Sigma}_i$. More details and discussions of GMM can be found in a review article (McNicholas 2016).

The mixtures of factor analyzers (MFA) model is a special case of GMM where the covariance matrix of each cluster is assumed to have the form of a factor analysis model, i.e., $\boldsymbol{\Sigma}_i = \boldsymbol{\Lambda}_i \boldsymbol{\Lambda}_i' + \boldsymbol{\Psi}_i$, where the $\boldsymbol{\Psi}_i$ are diagonal matrices and the $\boldsymbol{\Lambda}_i$ are $p \times q$ matrices that give the loading of the p observed variables onto q latent factors for the i th cluster. Compared to the general GMM, there are fewer parameters in the MFA to be estimated when $p \gg q$. For example, if there are $G = 3$ clusters and $p = 100$ features, then the general GMM will have 15,150 free parameters. The unconstrained MFA model with $q = 10$ factors will have only 3165 free parameters. As a result, the computation required will be reduced. In addition, if the latent variable representation of each cluster is properly specified, then the results from MFA will be as good as the general GMM. There are variants for the MFA model. In Ghahramani et al. (1996), the variance is modeled as $\boldsymbol{\Sigma}_i = \boldsymbol{\Lambda}_i \boldsymbol{\Lambda}_i' + \boldsymbol{\Psi}$; Tipping and Bishop (1999) proposed the mixture of probabilistic principal component analyzers model with $\boldsymbol{\Sigma}_i = \boldsymbol{\Lambda}_i \boldsymbol{\Lambda}_i' + \boldsymbol{\Psi}_i \mathbf{I}$; McLachlan and Peel (2000) proposed a more general model with $\boldsymbol{\Sigma}_i = \boldsymbol{\Lambda}_i \boldsymbol{\Lambda}_i' + \boldsymbol{\Psi}_i$. Each of these has a different degree of assumption about the similarity and complexity of the cluster variance structures, which can greatly affect the clustering results. These constraints allow for a greater or less degree of parsimony in the MFA models. McNicholas and Murphy (2008) extended these variants together into a family of eight Gaussian mixture models (called parsimonious Gaussian mixture models, or PGMM) by imposing the constraints described in Table 1.

Table 1 Eight Gaussian mixture models of the PGMM family where p is the number of covariates, q is the number of factors within each cluster, and G is the number of clusters

$\boldsymbol{\Lambda}_g = \boldsymbol{\Lambda}$	$\boldsymbol{\Psi}_g = \boldsymbol{\Psi}$	$\boldsymbol{\Psi}_g = \boldsymbol{\Psi}_g \mathbf{I}_p$	Covariance structure	Free parameters
C	C	C	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda} \boldsymbol{\Lambda}' + \boldsymbol{\Psi} \mathbf{I}_p$	$[pq - q(q-1)/2] + 1$
C	C	U	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda} \boldsymbol{\Lambda}' + \boldsymbol{\Psi}$	$[pq - q(q-1)/2] + p$
C	U	C	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda} \boldsymbol{\Lambda}' + \boldsymbol{\Psi}_g \mathbf{I}_p$	$[pq - q(q-1)/2] + G$
C	U	U	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda} \boldsymbol{\Lambda}' + \boldsymbol{\Psi}_g$	$[pq - q(q-1)/2] + Gp$
U	C	C	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda}_g \boldsymbol{\Lambda}_g' + \boldsymbol{\Psi} \mathbf{I}_p$	$G[pq - q(q-1)/2] + 1$
U	C	U	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda}_g \boldsymbol{\Lambda}_g' + \boldsymbol{\Psi}$	$G[pq - q(q-1)/2] + p$
U	U	C	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda}_g \boldsymbol{\Lambda}_g' + \boldsymbol{\Psi}_g \mathbf{I}_p$	$G[pq - q(q-1)/2] + G$
U	U	U	$\boldsymbol{\Sigma}_g = \boldsymbol{\Lambda}_g \boldsymbol{\Lambda}_g' + \boldsymbol{\Psi}_g$	$G[pq - q(q-1)/2] + Gp$

C, constrained; U, unconstrained

The alternating expectation conditional maximization (AECM; Meng and Dyk 1997), a variant of the EM algorithm, is commonly used for parameter estimation of the PGMM. Details can be found, for instance, from Ghahramani et al. (1996), McNicholas and Murphy (2008), and McLachlan et al. (2003). There are also Bayesian implementations of MFA implementation where MCMC is utilized for parameter estimation (Utsugi and Kumagai 2001; Lopes and West 2004; Fokoué and Titterton 2003). Infinite mixtures of infinite factor analyzers is a Bayesian method that simultaneously infers cluster-specific numbers of factors and the number of clusters (Murphy et al. 2020).

We developed Bayesian implementations of the eight PGMM variance structures based on Utsugi and Kumagai (2001) along with a procedure for inference on the best variance structure and number of components. Section 2 sets out the notation for the MFA. Then, in Sections 3 and 4, we present new methods for fully Bayesian inference on the number of groups and the choice of covariance constraints in an MFA. Section 5 combines the techniques in the previous two sections, performing inference in both dimensions simultaneously. Simulations are shown in Section 6 while Section 7 applies our new method to classifying tumor types from DNA methylation. Two benchmark datasets are analyzed in the Supplemental Material A.5.

2 A Bayesian Mixture of Factor Analysers

Here, we summarize the MCMC estimation of the fully unconstrained MFA by Utsugi and Kumagai (2001). The details of the priors and posteriors for the unconstrained MFA are shown in the Supplemental Material A.1.

Let n be the sample size, p be the number of features, and m be the number of clusters; $\mathbf{X}(p \times n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)$ be the observation matrix of continuous values; $q_k (k = 1, \dots, m)$ be the number of factors in cluster k ; $\mathbf{Y} = \{\mathbf{Y}_k(q_k \times n) = (\mathbf{y}_{k1}, \dots, \mathbf{y}_{kn}), k = 1, \dots, m\}$ be the factor scores; $\mathbf{Z}(m \times n) = (\mathbf{z}_{k1}, \dots, \mathbf{z}_{kn})$ be the cluster indicator (or allocation variable); $\tau = (\tau_1, \dots, \tau_m)$ be the prior selection probabilities; $\mathbf{M} = \{\mu_k(p \times 1), k = 1, \dots, m\}$ be the centroids of each cluster; $\mathbf{\Lambda} = \{\mathbf{\Lambda}_k(p \times q_k) = (\mathbf{\Lambda}_{k1}, \dots, \mathbf{\Lambda}_{kq_k}), k = 1, \dots, m\}$ be the loading matrix; and $\mathbf{\Psi} = \{\mathbf{\Psi}_k(p \times p) = \text{diag}(\mathbf{\Psi}_{k1}, \dots, \mathbf{\Psi}_{kp}), k = 1, \dots, m\}$ be the error matrix. Also, we write $\Theta = \{\tau, \mathbf{M}, \mathbf{\Lambda}, \mathbf{\Psi}\}$.

The values of q and m and the variables used in the model either need to be inferred, selected using model selection criteria or specified by the user. In current implementations of the MFA model, these decisions are made by comparing model fit across a limited set of combinations of values. We consider q to be a tuning parameter and choose it to be much smaller than the number of variables.

In this model, the likelihood is:

$$f(\mathbf{X}|\Theta, \mathbf{Y}, \mathbf{Z}) = \prod_{i=1}^n \prod_{k=1}^m [N(\mathbf{x}_i | \mathbf{\Lambda}_k \mathbf{y}_{ki} + \mu_k, \mathbf{\Psi}_k)]^{\mathbf{z}_{ki}}.$$

3 Inferring the Number of Clusters in a Mixtures of Factor Analysers Model

In clustering models, there may be several interpretable partitions. For example, there may be a coarse clustering into two groups and a more fine clustering into six smaller groups.

The model fit for each of these situations may be similar, but one will always have the higher likelihood. Additionally, the number of parameters in six groups is many more than the number in two groups and model selection methods that reward parsimony will generally choose the smaller models.

Since the dimensions and interpretation of the other mixture parameters (group means, group proportions, etc.) depend on the number of clusters, m , estimation methods for mixture models are usually conditional on a fixed m . These can be fit for a sequence of different m and then compared by model fit. However, this does not give fully Bayesian joint inference about the relative support for different values of m and learn the corresponding group parameters across models.

There is previous literature allowing m to be unknown. Escobar and West (1995) proposed a Bayesian clustering model using Dirichlet priors and considering the problem as “Bayesian nonparametrics”. Mengersen and Robert (1996) used a Kullback-Leibler distance and Roeder and Wasserman (1997) used a Schwarz criterion to select the unknown number of components. Nobile (1994), Phillips and Smith (1996), and Richardson and Green (1997) estimated the unknown m in a GMM with a fully Bayesian model.

We propose an MFA model allowing the number of components m to be variable. The approach is similar to Richardson and Green (1997) where they use RJMCMC (reversible-jump Markov chain Monte Carlo) to choose the number of components in a general GMM.

3.1 Model Specification

We use the priors and full conditional distributions from Section A.1 without fixing the number of clusters m . The target distribution for our RJMCMC sampler is the joint posterior distribution, π . It has the same form for any value of m , but the dimension of the parameters and priors changes. When comparing the posterior distribution for two different values of m , we call the distribution on the smaller dimension π_0 and on the larger dimension π_1 .

3.2 RJMCMC Procedure

To estimate the MFA with unknown m , we sample from an MCMC sampler for the posterior distribution. At each iteration, we sample values for each of the model parameters and the empirical distribution of these is an estimate of the joint posterior distribution. Because the dimensions of the parameters will change, we use a reversible jump MCMC (RJMCMC) sampler which can change in a single iteration by adding or removing a cluster. The two ways to add a cluster are called split and birth, meaning that you split an existing cluster or that you create a brand new empty cluster. The two ways to remove a cluster are called combine and death, meaning that you combine two existing clusters or that you remove an empty cluster. When the number of clusters does not change, the move is called stay and it simplifies to a standard MCMC update of the MFA model.

Let the current number of clusters be m_{curr} , the current values of the parameters are $\Theta_{\text{curr}} = \{\tau_{\text{curr}}, \mathbf{M}_{\text{curr}}, \mathbf{A}_{\text{curr}}, \mathbf{\Psi}_{\text{curr}}\}$, $\mathbf{Y}_{\text{curr}}$ and $\mathbf{Z}_{\text{curr}}$. The details of the five move types {stay, birth, death, split, combine} are as follows:

Stay: we do not attempt to change the number of clusters m_{curr} and just update the other parameters by Gibbs sampling using posterior distributions from Section A.1.

Birth: we propose an empty cluster. We first generate proposal parameters $\Theta_{\text{new}} = \{\tau_{\text{new}}, \mathbf{M}_{\text{new}}, \mathbf{\Lambda}_{\text{new}}, \mathbf{\Psi}_{\text{new}}\}$ and $\mathbf{Y}_{\text{new}}$ from equations to be given in Section 3.3, leaving the $\mathbf{Z}_{\text{curr}}$ unchanged and then compute the acceptance probability:

$$\alpha = \min \left\{ 1, \frac{\pi_1(\Theta_{\text{curr}}, \mathbf{Y}_{\text{curr}}, \mathbf{Z}_{\text{curr}}, \Theta_{\text{new}}, \mathbf{Y}_{\text{new}})}{\pi_0(\Theta_{\text{curr}}, \mathbf{Y}_{\text{curr}}, \mathbf{Z}_{\text{curr}})\phi(\Theta_{\text{new}}, \mathbf{Y}_{\text{new}})} \right\},$$

where π_0, π_1 are the density functions for old model and new model and ϕ is the proposal function.

Death: is the reversible move of the “birth” move. We first randomly choose an empty cluster, say cluster k and then drop $\Theta_k = \{\tau_k, \mathbf{M}_k, \mathbf{\Lambda}_k, \mathbf{\Psi}_k\}$ and $\mathbf{Y}_k$, leaving $\mathbf{Z}_{\text{curr}}$ unchanged as cluster k is an empty cluster. The acceptance probability is:

$$\beta = \min \left\{ 1, \frac{\pi_0(\Theta_{\text{curr} \setminus k}, \mathbf{Y}_{\text{curr} \setminus k}, \mathbf{Z}_{\text{curr}})\phi(\Theta_k, \mathbf{Y}_k)}{\pi_1(\Theta_{\text{curr} \setminus k}, \mathbf{Y}_{\text{curr} \setminus k}, \mathbf{Z}_{\text{curr}}, \Theta_k, \mathbf{Y}_k)} \right\}.$$

Combine: Our combine move begins by choosing a pair of components (j_1, j_2) at random, with weight equal to the reciprocal of the number of subjects within each component. When (j_1, j_2) are merged, the number of clusters, m , is reduced by 1. This forms a new component, here labeled j^* , we reallocate all those observations in (j_1, j_2) to j^* and create values for $(\Theta_{j^*}, \mathbf{Y}_{j^*}, \mathbf{Z}_{j^*})$. The $(\mathbf{\Lambda}_{j^*}, \mathbf{\Psi}_{j^*})$ are proposed from their prior distributions ((2) and (3)). For (τ_{j^*}, μ_{j^*}) , we set

$$\tau_{j^*} = \tau_{j_1} + \tau_{j_2}$$

$$\tau_{j^*}\mu_{j^*} = \tau_{j_1}\mu_{j_1} + \tau_{j_2}\mu_{j_2}$$

Finally, the $\mathbf{Y}_{j^*}$ are proposed by (5). The probability of accepting this combine is

$$\beta = \min \left\{ 1, \frac{\pi_0(\Theta_{\text{curr} \setminus \{j_1, j_2\}}, \mathbf{Y}_{\text{curr} \setminus \{j_1, j_2\}}, \mathbf{Z}_{\text{curr} \setminus \{j_1, j_2\}}, \Theta_{j^*}, \mathbf{Y}_{j^*}, \mathbf{Z}_{j^*})\phi_2(\Theta_{j_1, j_2}, \mathbf{Y}_{j_1, j_2}, \mathbf{Z}_{j_1, j_2})}{\pi_1(\Theta_{\text{curr} \setminus j^*}, \mathbf{Y}_{\text{curr} \setminus j^*}, \mathbf{Z}_{\text{curr} \setminus j^*}, \Theta_{j_1, j_2}, \mathbf{Y}_{j_1, j_2}, \mathbf{Z}_{j_1, j_2})\phi_1(\Theta_{j^*}, \mathbf{Y}_{j^*}, \mathbf{Z}_{j^*})} \right\}.$$

Split: The split move is the reversible move of the combine move. A component j^* is chosen at random with weight equal to the number of subjects within each components. Then it is split into two, labeled j_1 and j_2 . We first propose $(\mathbf{\Lambda}_{j_1}, \mathbf{\Lambda}_{j_2}, \mathbf{\Psi}_{j_1}, \mathbf{\Psi}_{j_2})$ from their prior distributions ((2) and (3)). The parameters for j_1 and j_2 must conform to the combine step. To achieve this, there is 1 degree of freedom for (τ_{j_1}, τ_{j_2}) , and p degrees of freedom for (μ_{j_1}, μ_{j_2}) so we use beta and gamma distributions to generate a_1 and a_2 to specify the new parameters.

$$a_1 \sim \text{beta}(2, 2) \quad \text{and} \quad a_{2i} \sim \text{Gamma}(1, 2), i = 1, 2, \dots, p.$$

Then, we set

$$\tau_{j_1} = \tau_{j^*}a_1, \quad \tau_{j_2} = \tau_{j^*}(1 - a_1),$$

$$\mu_{j_1i} = \mu_{j^*i} - a_{2i} \mathbf{\Sigma}_{j^*i} \sqrt{\left(\frac{\tau_{j_2}}{\tau_{j_1}}\right)}, i = 1, 2, \dots, p,$$

$$\mu_{j_2i} = \mu_{j^*i} + a_{2i} \mathbf{\Sigma}_{j^*i} \sqrt{\left(\frac{\tau_{j_1}}{\tau_{j_2}}\right)}, i = 1, 2, \dots, p,$$

where Σ_{j^*i} are the squared diagonal elements of $\Psi_{j^*} + \Lambda_{j^*}\Lambda'_{j^*}$. Finally, the $\mathbf{Y}_{j_1}, \mathbf{Y}_{j_2}$ are proposed by (5) and the reallocation is proposed by (4) and (6). The probability of accepting the split is

$$\alpha = \min \left\{ 1, \frac{\pi_1(\Theta_{\text{curr} \setminus j^*}, \mathbf{Y}_{\text{curr} \setminus j^*}, \mathbf{Z}_{\text{curr} \setminus j^*}, \Theta_{j_1, j_2}, \mathbf{Y}_{j_1, j_2}, \mathbf{Z}_{j_1, j_2}) \phi_1(\Theta_{j^*}, \mathbf{Y}_{j^*}, \mathbf{Z}_{j^*})}{\pi_0(\Theta_{\text{curr} \setminus j_1, j_2}, \mathbf{Y}_{\text{curr} \setminus j_1, j_2}, \mathbf{Z}_{\text{curr} \setminus j_1, j_2}, \Theta_{j^*}, \mathbf{Y}_{j^*}, \mathbf{Z}_{j^*}) \phi_2(\Theta_{j_1, j_2}, \mathbf{Y}_{j_1, j_2}, \mathbf{Z}_{j_1, j_2})} \right\}.$$

Notice that the birth and death moves are optional; however, adding them in practice will improve the efficiency of the algorithm as their acceptance rates are generally higher than that of split and combine moves.

3.3 Birth Proposal Distributions

To add a new empty cluster to the current model, we need proposal distributions to generate parameters. Concretely, we generate $\Theta_{\text{new}} = \{\tau_{\text{new}}, \mathbf{M}_{\text{new}}, \Lambda_{\text{new}}, \Psi_{\text{new}}\}$ and $\mathbf{Y}_{\text{new}}$ from ϕ , the product of five independent proposal distributions. The proposal distributions are specified as follows:

- $\tau_{\text{new}} \sim \text{Beta}(1, m_{\text{curr}})$, where m_{curr} is the current number of clusters. The τ vector is rescaled when τ_{new} is added: $\tau = \tau_{\text{old}} / 1 - \tau_{\text{new}}$ so that the new vector sums to 1.
- $\Psi_{\text{new}} = \text{diag}(\Psi_{\text{new},1}, \dots, \Psi_{\text{new},p})$, for each $j \in \{1, 2 \dots p\}$, $\Psi_{\text{new},j} \sim \text{IG}(\delta, \beta)$, where IG stands for inverse-gamma.
- $\mathbf{M}_{\text{new}} \sim N(\bar{\mathbf{x}}, \tilde{\alpha}_1^{-1} \Psi_{\text{new}})$, where $\bar{\mathbf{x}}$ is the mean of scaled dataset $\mathbf{X}$.
- $\Lambda_{\text{new}} = (\Lambda_{\text{new},1}, \dots, \Lambda_{\text{new},q_{\text{new}}})$, for each $j \in \{1, 2 \dots q_{\text{new}}\}$, $\Lambda_{\text{new},j} \sim N(\mathbf{0}, \tilde{\alpha}_2^{-1} \Psi_{\text{new}})$.
- $\mathbf{Y}_{\text{new}} = (y_{\text{new},1}, \dots, y_{\text{new},n})$, for each $j \in \{1, 2 \dots n\}$, $y_{\text{new},j} \sim N(\mathbf{0}, \mathbf{I}_{q_{\text{new}}})$.

The tuning parameters of the proposal distributions are $(\delta, \beta, \tilde{\alpha}_1, \tilde{\alpha}_2)$. They can be adjusted to change the RJMCMC acceptance rate.

3.4 Overfitting Bayesian MFA

Panagiotis (2018) shows that for an overfitted mixture model (where the true number of clusters, K_0 , is smaller than the number of fitted clusters, K), the asymptotic behavior of the posterior distribution of the $K - K_0$ redundant components depends on the prior distribution of mixing proportions (w). Let d denote the dimension of free parameters of the emission distribution. For the case of a Dirichlet prior distribution in (1), if $\max\{\gamma_k; k = 1, \dots, K\} < d/2$ then the posterior weight of the extra components converges to zero (Theorem 1 of Rousseau and Mengersen (2011)).

Using this theorem, there is a version of our algorithms where we initialize m with a relatively large value and only use **birth** and **death** steps. In this way, the computation as well as running time may be reduced; we compare this algorithm to the one using all four update steps in the simulation section.

4 Inferring the Covariance Structure in a Mixtures of Factor Analyzers Model

Though MFA models are more parsimonious than GMM in many cases, they can still have a large number of parameters. This can hurt the estimation of cluster variances, particularly

for smaller clusters. One way to further reduce the number of parameters for each cluster is to impose constraints (assumptions) on the covariance structures. In this section, we propose a fully Bayesian model for posterior inference on the covariance structures.

The MFA model can differ in whether the loading matrices are identical across clusters ($\Lambda_g = \Lambda$), whether the covariance of the noise is identical across clusters ($\Psi_g = \Psi$) and whether or not the noise covariance is constant for all variables ($\Psi_g = \Psi_g \mathbf{I}_p$). We use three letters (U or C) to represent the covariance constraints. The eight possible constraint combinations are listed in Table 1.

4.1 Model Specification

Inference on the MFA model with constrained covariance matrices is performed using similar inferential techniques to those outlined in Section 3 when inferring the number of clusters. The difference is that we fix the number of clusters, m , while allowing the covariance constraints to be unfixed. In this model, the dimensions of Ψ_k and Λ_k change as the constraint changes. The priors for z_i , τ , and μ_k remain the same as described earlier. To obtain inference on the three covariance constraints, they are each given a prior where unconstrained (U) and constrained (C) are equally likely.

$$C_c \sim \text{DiscreteUnif}(U, C), \quad c = 1, 2, 3.$$

The priors for Ψ_k and Λ_k are shown as follows:

For $\{\Psi_k = \text{diag}(\Psi_{k1}, \dots, \Psi_{kp}), k = 1, \dots, m\}$:

If $\Psi_k = \Psi$ ($C_2 = C$), for $k = 1, 2, \dots, m$, $\Psi_j \sim IG(\delta, \beta)$, $j = 1, 2, \dots, p$.

If $\Psi_k = \Psi_k \mathbf{I}_p$ ($C_3 = C$), $\Psi_k \sim IG(\delta, \beta)$, $k = 1, 2, \dots, m$.

For $\Lambda = \{\Lambda_k = (\Lambda_{k1}, \dots, \Lambda_{kq_k}), k = 1, \dots, m\}$:

If $\Lambda_k = \Lambda$ and $\Psi_k = \Psi$ ($C_1 = C$ and $C_2 = C$), $k = 1, 2, \dots, m$, $\Lambda_j \sim N(0, \alpha_2^{-1} \Psi)$, $j = 1, 2, \dots, q_k$.

If $\Lambda_k = \Lambda$ and $\Psi_k = \Psi_k$ ($C_1 = C$ and $C_2 = U$), $k = 1, 2, \dots, m$, $\Lambda_j \sim N(0, \alpha_2^{-1} \left(\frac{1}{m} \sum_{k=1}^m \Psi_k^{-1} \right)^{-1})$, $j = 1, 2, \dots, q_k$.

Combining these with the likelihood, we get the joint posterior distribution, π , which is the target distribution for our RJMCMC sampler. The posterior distribution has a different form for different covariance constraints, because the dimension of the parameters Ψ and Λ and the form of their priors change. When comparing the posterior for two different values of C_c in an RJMCMC step, we call the distribution without the constraint π_0 and that with the constraint π_1 .

4.2 Conditional Posterior Distributions Under Different Covariance Constraints

The posteriors for parameters Y , Z , τ , and M derived in Section A.1 are the same for all eight covariance constraints, while the posteriors for Λ and Ψ are different under each constraint.

The unconstrained log posterior density function of Λ_k and Ψ_k can be expressed as follow:

$$\log f(\Lambda_k | \mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{H}) = \sum_{k=1}^m \left\{ -\frac{1}{2} \text{tr} [(-2C_{\mathbf{X}_\mu \mathbf{Y} k} \Lambda_k') \Psi_k^{-1}] \right. \\ \left. - \frac{1}{2} \text{tr} [(C_{\mathbf{Y} \mathbf{Y} k} + \alpha_2 \mathbf{1}_{q_k}') (\Lambda_k' \Psi_k^{-1} \Lambda_k)] \right\} + C_1,$$

$$\begin{aligned} \log f(\Psi_k | \mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{H}) = & \sum_{k=1}^m \left\{ \frac{1}{2} [n_k + q_k + (2\delta - 2) + 1] \log |\Psi_k^{-1}| \right. \\ & - \frac{1}{2} \text{tr} \left[(C_{XXk} - 2C_{X\tilde{Y}k} \tilde{\Lambda}'_k + 2\beta \mathbf{I}) \Psi_k^{-1} \right] \\ & \left. - \frac{1}{2} \text{tr} \left[(C_{\tilde{Y}\tilde{Y}k} + A_k) (\tilde{\Lambda}'_k \Psi_k^{-1} \tilde{\Lambda}_k) \right] \right\} + C_2, \end{aligned}$$

where C_1 and C_2 are constants and

$$\tilde{\Lambda}_k = (\mu_k, \Lambda_k), A_k = \text{diag}(\alpha_1, \alpha_2 \mathbf{1}'_{q_k}), q = \sum_{k=1}^m q_k, n_k = \sum_{i=1}^n z_{ki}.$$

Let

$$\mathbf{X}_\Lambda = x_i - \Lambda_k y_{ki}, \mathbf{X}_\mu = x_i - \mu_k, \tilde{y}_{ki} = c(1, y'_{ki}),$$

$$C_{XXk} = \sum_{i=1}^n z_{ki} x_i x'_i, C_{XYk} = \sum_{i=1}^n z_{ki} x_i y'_{ki}, C_{X1k} = \sum_{i=1}^n z_{ki} x_i, C_{YYk} = \sum_{i=1}^n z_{ki} y_{ki} y'_{ki}.$$

We derive the posterior of Λ and Ψ for the eight models using the properties of the multivariate normal distribution; the details are included in the supplemental materials.

4.3 RJMCMC Procedure

Similar to the estimation of the model for an unknown number of clusters, m , this estimation also uses RJMCMC. However, the acceptance rate under a simple choice of proposal distributions was extremely low. We use a multi-step proposal distribution in order to improve the rate of convergence of the sampler to the posterior distribution.

4.4 Multi-step Proposal Distribution

The parameters Λ and Y depend on each other, meaning a direct jump only for Λ will result in vast decrease of the density function, making the acceptance probability very small and slowing the convergence rate of the sampler. Larjo and Lähdesmäki (2015) developed a method to improve the efficiency of convergence for MCMC samplers. Using their notation, let $Q(k|i)$ be a proposal distribution from state i to k that can be decomposed into t independent distributions $Q_j, j = 1, \dots, t$, and $r = (r_1, r_2, \dots, r_{t-1})$ is a vector of intermediate state so that the whole move is $i \rightarrow r_1 \rightarrow \dots \rightarrow r_{t-1} \rightarrow k$. The probability of proposing the move is

$$Q(k|i) = Q_1(r_1|i) Q_2(r_2|r_1) \dots Q_t(k|r_{t-1}),$$

where $Q_j(r_j|r_{j-1})$ can be any functions for the move from r_{j-1} to r_j .

4.5 Updating the Variance Parameters

The update steps for the variance parameters are done separately for Λ and Ψ . The details for Ψ are shown in the Supplemental Materials A.3. We derive the acceptance probability

formulas for extension and contraction steps. For the extension step, that is, from $\Lambda \rightarrow \Lambda_k, k = 1, 2, \dots, m$:

1. propose $\Lambda_k, k = 1, 2, \dots, m$ with $T_1(\Lambda_k | \mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$, where T_1 is the conditional posterior of Λ_k given no constraint on Λ .
2. evaluate the reverse update $T_2(\Lambda | \mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$, where T_2 is the conditional posterior of Λ under the constraint on Λ .
3. compute the joint density of $\pi_0(\mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$.
4. update $(\Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$ using MCMC for s times with multi-step proposal, get $(\Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)$. The proposal functions of this step are

$$Q_1(\Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s | \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z}) \text{ and } Q_2(\Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z} | \Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s).$$

5. compute the joint density of $\pi_1(\mathbf{X}, \Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)$.

The acceptance probabilities can be calculated as follows:

$$\alpha = \min \left\{ 1, \frac{\pi_1(\mathbf{X}, \Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s) T_2(\Lambda | \mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})}{\pi_0(\mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z}) T_1(\Lambda_k | \mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})} \frac{Q_2(\Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z} | \Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)}{Q_1(\Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s | \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})} \right\}.$$

For the contraction step, that is $\Lambda_k (k = 1, 2, \dots, m) \rightarrow \Lambda$:

1. propose Λ with $T_2(\Lambda | \mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$.
2. evaluate the reverse update $T_1(\Lambda_k | \mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$.
3. compute the joint density of $\pi_1(\mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$.
4. update $(\Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$ using MCMC for s times with multi-step proposal, get $(\Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)$. The proposal functions of this step are

$$Q_1(\Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s | \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z}) \text{ and } Q_2(\Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z} | \Lambda_k^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s).$$

5. update $(\Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})$ using MCMC for s times, and get $(\Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)$.
6. compute the joint density of $\pi_0(\mathbf{X}, \Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)$.

The acceptance probabilities can be calculated as follows:

$$\beta = \min \left\{ 1, \frac{\pi_0(\mathbf{X}, \Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s) T_1(\Lambda_k | \mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})}{\pi_1(\mathbf{X}, \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z}) T_2(\Lambda | \mathbf{X}, \Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})} \frac{Q_1(\Lambda, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z} | \Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s)}{Q_2(\Lambda^s, \mathbf{M}^s, \Psi^s, \tau^s, \mathbf{Y}^s, \mathbf{Z}^s | \Lambda_k, \mathbf{M}, \Psi, \tau, \mathbf{Y}, \mathbf{Z})} \right\}.$$

Remark In practice, when we compute the acceptance probabilities, we assume Q_1 and Q_2 are symmetric proposal distributions. This makes the ratio cancel in the equations above. The goal of the assumption is to improve the efficiency of convergence. The effect of this assumption is that the within model state frequency may be changed while the between model frequency stays the same.

5 Inferring the Number of Clusters and Covariance Structure in a MFA Model

We proposed inferential procedures for two different model properties in the previous sections. If we are interested in both the number of clusters and the covariance structures, which model shall be used first? In fact, the result may be very different if we change the order as we only get the local optimum. In this section, we combine inference for the number of clusters and covariance structures so that the model can do both simultaneously.

5.1 Methodology

Our RJMCMC procedure comprises of the following steps:

1. update $\{\Theta, \mathbf{Y}, \mathbf{Z}\}$ from their full conditional.
2. propose an update for the number of clusters, accept with the probabilities above.
3. propose an update for the covariance structures, accept with the probabilities above.

One complete pass over these three steps will be called a sweep and is the basic time step of the algorithm.

6 Simulations for Model Inference in MFA

For each procedure introduced, we created a simulation study. We compared our method to existing methods based on inference for the parameters of interest and the adjusted Rand index (ARI; Hubert and Arabie 1985) of classification agreement with the truth.

6.1 Simulation for the Number of Clusters

For the variance structure, we simulated data from the MFA model using the **simData** function from the **fabMix** R package (Papastamoulis 2020). We used the default CCU model. Also, we set the sample size, $n = 500$, the number of variables, $p = 10$, the number of factors, $q = 4$, the true number of clusters, $k = 10$, and the common $\Psi = \text{diag}\{1/1, 1/2, \dots, 1/p\}$. In addition, we used the default setting for other parameters. The pairwise scatter plot for one example dataset is shown in Fig. 1.

Parameter Settings We set the tuning parameters of the proposal distributions and fixed the number of factors at 4. We ran a simulation study 100 times and for each simulation, the number of RJMCMC iterations was 2000 with 500 burn in iterations. The allowed range of the number of clusters was from 1–20. We fixed the sampler at the true constraint (CCU) and fit our model and the pgmm package (McNicholas et al. 2019) to choose the number of clusters.

Results We ran the simulation 100 times. We are interested in how our method chooses the number of clusters; see Table 2. In the two step (birth/death) case, it chooses 10 groups most often (45/100) with average ARI 0.929. In the four step (birth/death/split/combine) case, it equally chooses nine as well as 10 groups most often (37/100 each) with average ARI 0.908. For the same data, the pgmm package chooses 11 groups most often (31/100 times) with average ARI 0.877, meaning our inferential approach performs optimally in uncovering the clustering structure.

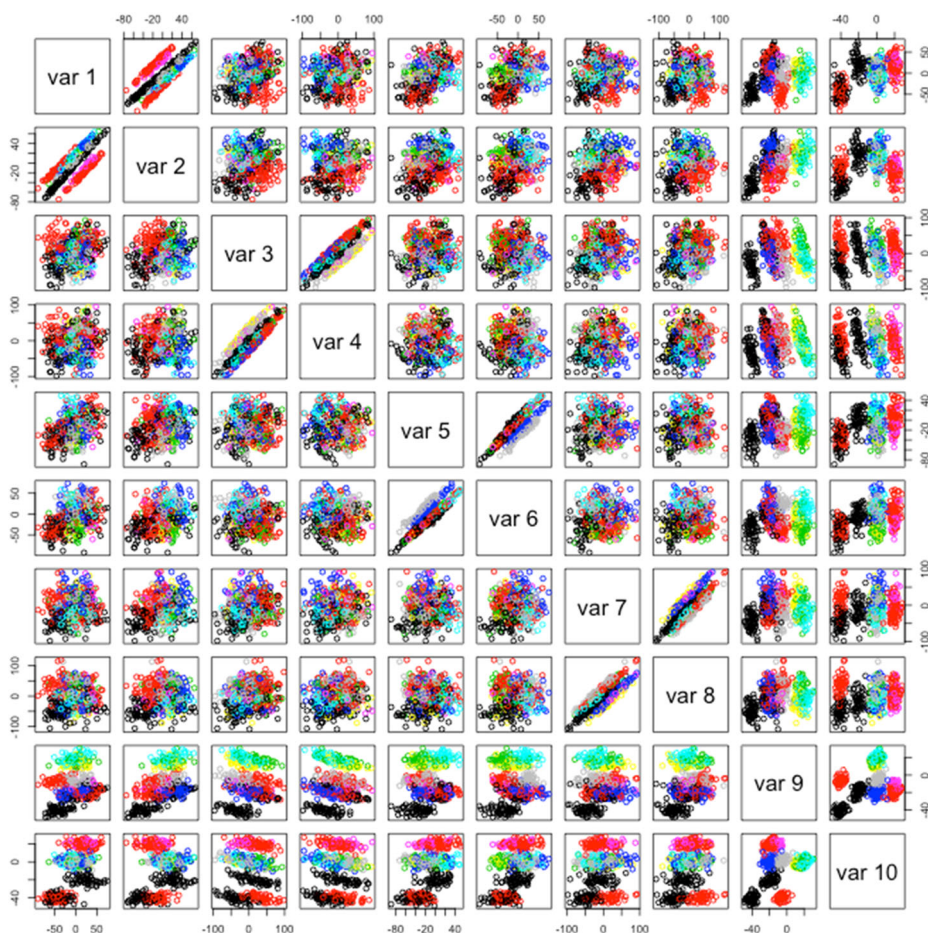


Fig. 1 Scatter plots of a simulated dataset with the number of clusters $k = 10$

6.2 Simulation for Choosing the Variance Structure

For these simulation studies, we take the number of clusters to be known and we generate data from each of the eight variance constraint models.

Table 2 For 100 simulations with 10 clusters, the distribution of the estimated number of clusters when using our method implemented with birth/death (B/D) or birth/death/split/combine (B/D/S/C) steps and the the pgmm method

	Number of clusters												
Method	6	7	8	9	10	11	12	13	14	15	16	17	18
B/D	0	2	10	41	45	0	0	1	0	0	0	0	1
B/D/S/C	1	6	19	37	37	0	0	0	0	0	0	0	0
pgmm	0	1	1	8	20	31	19	6	7	6	1	0	0

Bold entries demonstrate the performance of each model

To do this, we build datasets from a mixture of three factor analyzers model. For the unconstrained model,

$$\mathbf{X}_1 = \mu_1 + \mathbf{Y}_1 \mathbf{\Lambda}_1 + \epsilon_1, \quad \mathbf{X}_2 = \mu_2 + \mathbf{Y}_2 \mathbf{\Lambda}_2 + \epsilon_2, \quad \mathbf{X}_3 = \mu_3 + \mathbf{Y}_3 \mathbf{\Lambda}_3 + \epsilon_3,$$

where $\mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3 \sim N(0, \mathbf{I}_{10})$ and $\epsilon_1 \sim N(0, \Psi_1)$, $\epsilon_2 \sim N(0, \Psi_2)$, $\epsilon_3 \sim N(0, \Psi_3)$. For constrained models, if the first constraint is true, then we set $\mathbf{\Lambda}_3 = \mathbf{\Lambda}_2 = \mathbf{\Lambda}_1$; if the second constraint is true, we set $\Psi_3 = \Psi_2 = \Psi_1$; if the third constraint is true, we set $\Psi_1 = \Psi_1^*$, $\Psi_2 = \Psi_2^*$ and $\Psi_3 = \Psi_3^*$.

The simulation parameters are given in the Supplemental Material (A.4). Figure 2 shows a pairwise scatter plot of the first five dimensions for a single dataset under the UUU model.

Parameter Settings We set the tuning parameters of the proposal distributions and fixed the number of clusters at three and the number of factors at three. For each constraint, we ran a simulation study 100 times and for each simulation, the number of RJMCMC iterations is 5000. We initialized each sampler at the fully constrained model (CCC) and counted the times that our model and the pgmm package choose the true model. To assess convergence, the multivariate effective sample size (multiESS) was calculated for our method in each simulation (Vats et al. 2019). The parameters used for multiESS were the means of the clusters. The multiESS was at least 4000 in each simulation (see Table 3 for averages), so we should have good estimates of these quantities. Trace plots of the cluster means verified this assessment that the RJMCMC was converged.

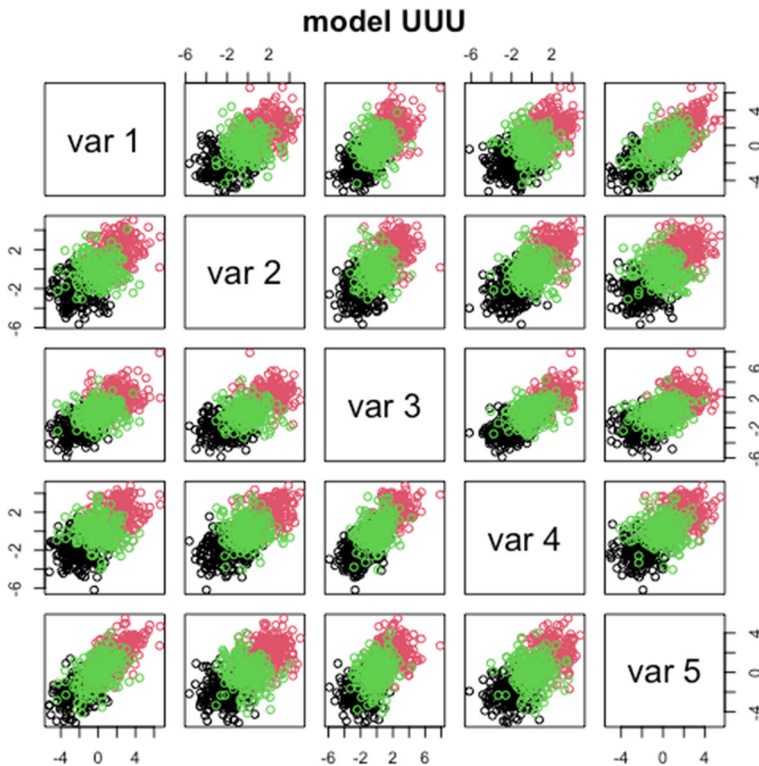


Fig. 2 Scatter plot of simulated data generated by the constraint UUU

Table 3 A simulation comparing bpgmm with pgmm for finding the correct covariance structure

	True model	% true model		Average ARI		Average ESS
		bpgmm	pgmm	bpgmm	pgmm	
1	(UUU)	100	9	0.86	0.93	7677
2	(UUC)	95	25	0.58	0.83	4341
3	(UCU)	81	80	0.97	0.90	8280
4	(UCC)	89	23	0.77	0.49	4421
5	(CUU)	99	20	0.94	0.96	4887
6	(CUC)	88	34	0.80	0.72	4394
7	(CCU)	98	52	0.97	0.66	4845
8	(CCC)	84	34	0.87	0.25	4173

Simulation Results Table 3 shows the percentage of correct selection of the variance structure and computed the average ARI for our method. This shows that our method does better job at finding the correct models than the ACEM algorithm implemented by the pgmm package. One explanation for this result could be that the EM algorithm tends to stop at a local maximum while the RJMCMC will converge to the global maximum. The disadvantage of our RJMCMC algorithm is that it runs much slower than the EM algorithm. Another explanation could be that the penalties on complexity that the RJMCMC algorithm creates are different from those implemented by using the Bayesian information criterion (BIC; Schwarz et al. 1978) to compare different parameter dimensionality.

The ARI for the bpgmm and the pgmm are often quite similar with the bpgmm having higher average ARI in six of the eight cases. For the two most unconstrained variance structures (UUU and UUC), pgmm is better at finding the correct partition (as shown by the higher ARI), despite choosing a different (likely more parsimonious) set of variance constraints. This could be considered a feature of the pgmm: that it can recover the correct partition with fewer parameters. However, the most constrained model for the simulation (CCC) shows better performance for our method in both identifying the variance constraints and the partition.

7 A DNA Methylation Example for Model Selection in MFA

Precise diagnoses for cancer are crucial for selecting an accurate treatment plan for patients in personalized medicine. The cancer methylome is a combination of both somatically acquired DNA methylation changes and characteristics reflecting the cell of origin (Hoadley et al. 2018), which makes it suitable for tumor classification and stratification of cancer patients. In fact, methylation data-based tumor classification has been a leading topic in oncology (Rodríguez-Paredes and Manel 2011; Sturm et al. 2016; Capper et al. 2018). However, there is no single classification model that can perform model selection, feature selection, and classification at the same time for this type of dataset. Capper et al. (2018) have published a dataset with 2801 patients DNA-methylation samples for 91 labeled diagnostic categories of brain tumors. We have only used a subset of this large dataset. Specifically, the subset contains 576 observations and uses only the 20 most variable DNA methylation features and the five largest brain tumor categories from the cohort data. The DNA methylation data are between 0 and 1, so transformation might improve the normality

of these features. However, we are using them on the original scale. The pairwise scatter plots of the first 10 features are in Fig. 3. In this example, we use the procedures of Sections 3 and 4 jointly to let the model choose the number of clusters and the variance structures simultaneously.

Parameter Settings We set the tuning parameters of the proposal distributions, the cluster number range from 1–10, all eight possible covariance constraints, and fixed the number of factors at four. The number of RJMCMC iterations was 15,000 after 5000 burn-in iterations. Trace plots of the cluster means were used to diagnose convergence. We initialized the sampler at the fully unconstrained model (UUU) with $m = 10$ clusters.

Results Our model chooses five clusters almost always (14,028/15,000, see Table 4), meaning it chooses the correct number of clusters. The ARI for the maximum posterior clustering is 0.981, meaning the estimated clustering is very similar to the given brain cancer labels.

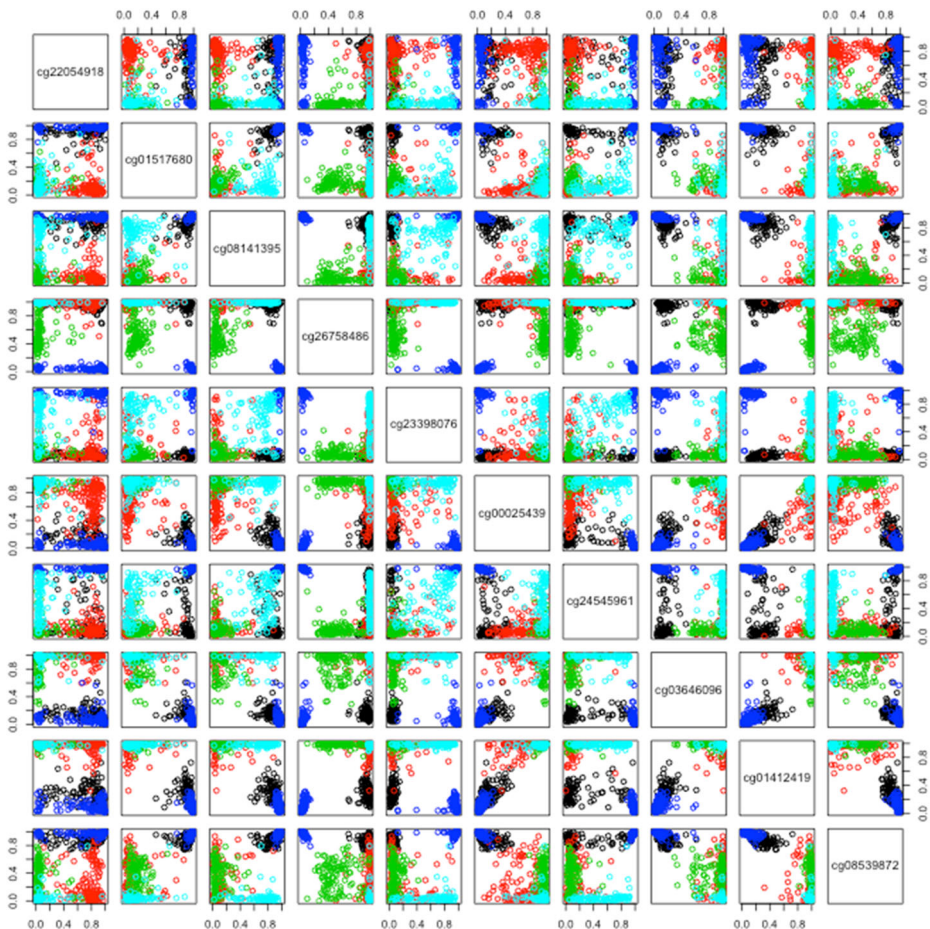


Fig. 3 Scatter plots of 10 DNA methylation features. The colors depict the five brain tumor labels

Table 4 Posterior distribution for the number of clusters from our method

Proportion of clusters	5	6
Number of iterations	93.5%	6.5%

Also, it always chooses the unconstrained variance structure UUU (15,000/15,000), meaning that the variance of the five clusters are not identical. Comparing our method with other existing methods, the *pgmm* package with the number of clusters ranged from one to nine chooses nine clusters and CUU as its best fit, with its ARI of 0.745. Another possible method is the *fabMix* package which uses a parallel tempered Gibbs sampler to estimate overfit MFA with the same eight variance constraints. The *fabMix* package, with a maximum number of clusters set at 10, chooses seven clusters and CUU as its best fit, with its ARI of 0.899. This demonstrates that our method outperforms the others.

A Couple of Benchmark Datasets In the Supplemental Material [A.5](#), we present two benchmark datasets: *wine* and *olive*. For each one, inference is done for both the number of clusters and the variance structure using the method described here. In each case, the inference for the number of clusters does not match the regional origin labels used for comparison. The *pgmm* package finds a partition with a higher ARI for those labels than the proposed method in both cases.

In each of these popular datasets, our RJMCMC sampler moves much more than in the DNA methylation example. The datasets are small and only a handful of iterations are needed to reach convergence, but several different numbers of clusters are explored as well as several variance constraints. These examples illustrate that while the RJMCMC does not move from the UUU model in the DNA methylation example and rarely moves away from $m = 5$ clusters, that is not a characteristic of the sampler, but of the data.

8 R Package *bpgmm*

As mentioned previously, our method out performs the estimation of the correct covariance constraints and number of clusters, compared to the method *pgmm* with uses AECM estimation and post-hoc model choice by BIC. However, our RJMCMC algorithm takes considerably longer to run. To address this, we developed an R package *bpgmm* (Bayesian parsimonious Gaussian mixture models; Lu et al. 2020) implementing the algorithm in this paper and providing it to other R users. We embedded C++ code inside R to improve the efficiency of the algorithm and, as expected, the version using C++ is much faster than the version using only R code. For example, in one simulation from Section 6.2 where data is generated from a fully unconstrained (UUU) model, drawing 5000 RJMCMC iterations took 4280 s for the R version and only 2738 s for the C++ version.

9 Discussion

The model for MFA embeds factor analysis into a Gaussian mixture model in order to estimate the variance structure more parsimoniously. The factor analysis model can extract a set of local linear subspaces hidden in data so the MFA model is particularly efficient at clustering when the number of variables is large and the variables can be represented as linear combination of factors. However, the RJMCMC methods used for estimation here do not scale well for large p .

Here, we developed an RJMCMC inferential procedure to infer the number of clusters and the covariance. To get fully Bayesian inference over the model choices, we use RJMCMC to select the best number of clusters and the best variance structures. We derived the posterior distributions of the variance parameters under all eight constraints in the PGMM family and used RJMCMC to do model selection among these different model structures. Simulation results showed that our method is much better at finding the correct variance structure than the algorithm in R package `pgmm` (McNicholas and Murphy 2008). That method selects a different variance constraint for the DNA methylation data example and too many clusters. However, in benchmark datasets shown in the Appendix, the `pgmm` package performed similarly or better than the one proposed here. Based on simulation results and the labeled brain cancers types, our method gives the more reliable grouping of patients.

Comparing with some other model selection methods, ours are fully Bayesian and model based. In addition, our method does clustering and model selection simultaneously; as a result, our algorithm may be better at finding the global maximum. Since clustering methods are data driven, our methods do not only offer a single solution but give the relative posterior data support for different possible results. This makes sense as there can be different explanations for different choices of the number of clusters, and the type of covariance constraints. However, in our simulations, there is only one correct partition. In the analysis of example data, our method never found posterior support for very different numbers of clusters. So, we did not find any evidence that multiple clustering solutions would be useful. The RJMCMC methods used for estimation here do not scale well for large p . We did not perform inference on the number of factors in the MFA model as was done in McNicholas and Murphy (2008). We did this in Lu (2019) and this method of inference does not appear well-suited to those model properties.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00357-021-09391-8>.

References

- Blake, C. (1998). Uci repository of machine learning databases. <https://archive.ics.uci.edu/ml/index.php>.
- Capper, D., Jones, D.T.W., Sill, M., Hovestadt, V., Schrimpf, D., Sturm, D., Koeilsche, C., Sahm, F., Chavez, L., Reuss, D.E., et al. (2018). DNA methylation-based classification of central nervous system tumours. *Nature*, 555(7697), 469–474.
- Diebolt, J., & Robert, C.P. (1994). Estimation of finite mixture distributions through Bayesian sampling. *Journal of the Royal Statistical Society: Series B (Methodological)*, 56(2), 363–375.
- Escobar, M.D., & West, M. (1995). Bayesian density estimation and inference using mixtures. *Journal of the american statistical association*, 90(430), 577–588.
- Fokoué, E., & Titterton, D.M. (2003). Mixtures of factor analysers. Bayesian estimation and inference by stochastic simulation. *Machine Learning*, 50(1-2), 73–94.
- Forina, M., Armanino, C., Lanteri, S., Tiscornia, E. (1983). Classification of olive oils from their fatty acid composition. In *Food research and data analysis: proceedings from the IUFOST Symposium, September 20-23, 1982, Oslo, Norway/edited by H. Martens and H. Russwurm, Jr, London, Applied Science Publishers*.
- Forina, M., Leardi, R., Armanino, C., Lanteri, S., Conti, P., Princi, P. (1988). PARVUS: An extendable package of programs for data exploration, classification and correlation. *Journal of Chemometrics*, 4(2), 191–193.
- Ghahramani, Z., Hinton, G.E., et al. (1996). The EM algorithm for mixtures of factor analyzers. Technical report, Technical Report CRG-TR-96-1 University of Toronto.

- Hoadley, K.A., Yau, C., Hinoue, T., Wolf, D.M., Lazar, A.J., Drill, E., Shen, R., Taylor, A.M., Cherniack, A.D., Thorsson, V. (2018). Cell-of-origin patterns dominate the molecular classification of 10,000 tumors from 33 types of cancer. *Cell*, 173(2), 291–304.
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of classification*, 2(1), 193–218.
- Larjo, A., & Lähdesmäki, H. (2015). Using multi-step proposal distribution for improved MCMC convergence in Bayesian network structure learning. *EURASIP Journal on Bioinformatics and Systems Biology*, 2015(1), 6.
- Lopes, H.F., & West, M. (2004). Bayesian model assessment in factor analysis. *Statistica Sinica*, 14(1), 41–67.
- Lu, X. (2019). Model selection and variable selection for the mixture of factor analyzers model. PhD thesis, University of Rochester.
- Lu, X., Li, Y., Love, T. (2020). bpgmm: Bayesian model selection approach for parsimonious Gaussian mixture models. URL <https://CRAN.R-project.org/package=bpgmm>. R package version 1.0.7.
- McLachlan, G., & Peel, D. (2000). Mixtures of factor analyzers. In *Proceedings of the seventeenth international conference on machine learning, San Francisco, pages 599–606. Morgan Kaufmann*.
- McLachlan, G.J., & Basford, K.E. (1988). *Mixture models: Inference and applications to clustering*. New York: Marcel Dekker Inc.
- McLachlan, G.J., Peel, D., Bean, R.W. (2003). Modelling high-dimensional data by mixtures of factor analyzers. *Computational Statistics & Data Analysis*, 41(3–4), 379–388.
- McNicholas, P.D. (2016). Model-based clustering. *Journal of Classification*, 33(3), 331–373.
- McNicholas, P.D., ElSherbiny, A., McDaid, A.F., Murphy, T.B. (2019). pgmm: Parsimonious Gaussian mixture models. <https://CRAN.R-project.org/package=pgmm>. R package version 1.2.4.
- McNicholas, P.D., & Murphy, T.B. (2008). Parsimonious Gaussian mixture models. *Statistics and Computing*, 18(3), 285–296.
- Meng, X.L., & Dyk, D.V. (1997). The EM algorithm—an old folk-song sung to a fast new tune. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 59(3), 511–567.
- Mengersen, K.L., & Robert, C.P. (1996). *Testing for mixtures: a Bayesian entropic approach*, MA: Oxford University Press, Cambridge.
- Murphy, K., Viroli, C., Gormley, I.C. (2020). Infinite mixtures of infinite factor analysers. *Bayesian Analysis*, 15(3), 937–963.
- Nobile, A. (1994). *Bayesian analysis of finite mixture distributions*. Pittsburgh: PhD thesis, PhD Thesis. Carnegie Mellon University.
- Panagiotis, P. (2018). Overfitting Bayesian mixtures of factor analyzers with an unknown number of components. *Computational Statistics & Data Analysis*, 124, 220–234.
- Papastamoulis, P. (2020). fabMix: Overfitting bayesian mixtures of factor analyzers with parsimonious covariance and unknown number of components. <https://CRAN.R-project.org/package=fabMix>. R package version 5.0.
- Phillips, D.B., & Smith, A.F.M. (1996). *Bayesian model comparison via jump diffusions*, (pp. 215–239). New York: Springer.
- Press, S.J., & Shigemasu, K. (1989). *Bayesian inference in factor analysis*, (pp. 271–287). New York: Springer.
- Richardson, S., & Green, P.J. (1997). On Bayesian analysis of mixtures with an unknown number of components (with discussion). *Journal of the Royal Statistical Society: series B (statistical methodology)*, 59(4), 731–792.
- Rodríguez-Paredes, M., & Manel, E. (2011). Cancer epigenetics reaches mainstream oncology. *Nature Medicine*, 17(3), 330.
- Roeder, K., & Wasserman, L. (1997). Practical Bayesian density estimation using mixtures of normals. *Journal of the American Statistical Association*, 92(439), 894–902.
- Rousseau, J., & Mengersen, K. (2011). Asymptotic behaviour of the posterior distribution in overfitted mixture models. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 73(5), 689–710.
- Schwarz, G. et al. (1978). Estimating the dimension of a model. *Annals of Statistics*, 6(2), 461–464.
- Sturm, D., Orr, B.A., Toprak, U.H., Hovestadt, V., Jones, D.T.W., Capper, D., Sill, M., Buchhalter, I., Northcott, P.A., Leis, I., et al. (2016). New brain tumor entities emerge from molecular classification of CNS-PNETs. *Cell*, 164(5), 1060–1072.
- Tipping, M.E., & Bishop, C.M. (1999). Mixtures of probabilistic principal component analyzers. *Neural Computation*, 11(2), 443–482.

- Utsugi, A., & Kumagai, T. (2001). Bayesian analysis of mixtures of factor analyzers. *Neural Computation*, 13(5), 993–1002.
- Vats, D., Flegal, J.M., Jones, G.L. (2019). Multivariate output analysis for Markov chain Monte Carlo. *Biometrika*, 106(2), 321–337.

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